

Dev Garg

☎ 9034729687 ✉ devg55030@gmail.com 🔗 linkedin.com/in/devgxxg 🌐 github.com/devgxxg 🌐 Portfolio

Experience

AI Engineer

March 2026 – Present

Dee Development Engineering Ltd (Dee Piping)

Palwal, Haryana (On-site)

- Engineered the Automated Inventory Code Allocation (AICD) system; a full-stack AI application (FastAPI, React, LLM) that dynamically matches complex MTO requirements against unallocated inventory using ASME B36.10/B36.19 logic, thickness interpolation, and step-down dimension algorithms.
- Building an end-to-end AI pipeline that extracts structured Bill of Materials (BOM) from complex isometric piping drawing PDFs using Claude Vision API, pdf2image, and openpyxl — processing multiple projects drawings.
- Developed an ICD Code Generator: automated 19-character piping component code lookup via LLM-based tool and deterministic master sheet querying, reducing hours of manual Excel work to seconds.

AI/ML Intern

June 2025 – August 2025

WESEE, Indian Navy

New Delhi (On-site)

- Built IMInsight — an open-source, offline-capable AI image analysis system using React and Flask to support naval officers in object detection, threat identification, and anomaly detection in defense surveillance imagery.
- Fine-tuned YOLOv8 and CNN-based models for high-accuracy detection of ships, personnel, radars, and aircraft; integrated CLIP and BLIP models to generate contextual image captions from surveillance feeds.
- Implemented NLP pipelines to convert visual outputs into structured, human-readable intelligence summaries for military-grade use cases.
- Designed and evaluated model performance across defense-specific edge cases including low-resolution satellite imagery and partially occluded targets, improving classification robustness for real-world naval scenarios.

Projects

GigScore | React, Vite, TailwindCSS, Flask, Python, GROQ AI, Pandas

Live

- Built an AI-powered alternative credit scoring platform for India's gig economy (Zomato riders, freelancers) — generates a dynamic GigScore (up to 850) by analyzing platform payouts, transaction trends, and expense ratios without relying on CIBIL scores.
- Integrated Groq AI Financial Coach that delivers personalized, actionable advice in English/Hinglish based on each worker's exact financial profile.
- Engineered a Python synthetic data engine generating 90-day realistic transaction profiles for multiple risk levels; presented at MasterX Hackathon.

BiteWise – AI Food Monitoring Assistant | Python, React, Flask, Gemini LLM, SQLite

Live

- Built a multimodal AI web app using Gemini LLM with vision to analyze food images — delivering health scores, allergen detection, and disease-risk insights via structured JSON responses.
- Engineered Flask backend with model fallback chain, retry logic, and prompt engineering for reliable inference; designed React frontend with real-time camera capture and animated health dashboards.
- Deployed via CI/CD pipelines — frontend on Netlify, backend on Render; maintained code quality with Pytest test suites and version control via Git.

IMInsight – Defense AI Image Analysis | Python, YOLOv8, BLIP, CLIP, Flask

Live

- Developed a defense-grade AI surveillance system enabling object detection, anomaly recognition, and multi-class classification in satellite and surveillance imagery.
- Fine-tuned YOLOv8 and CNN models for high-speed detection of ships, personnel, radars, and aircraft; integrated BLIP for NLP-based image captioning converting visual data into actionable intelligence.
- Designed modular Flask ML pipeline for API-driven image uploads and real-time inference, built to operate fully offline for secure defense environments.

HeartDiseaseDetective | Python, Scikit-learn, Pandas, Matplotlib

GitHub

- Built an end-to-end ML classification pipeline for heart disease risk prediction using clinical patient data including cholesterol, blood pressure, and ECG features.
- Evaluated multiple algorithms (Logistic Regression, Random Forest, SVM) with cross-validation; selected best-performing model based on precision, recall, and F1 metrics.
- Performed feature engineering and correlation analysis to identify key clinical predictors, improving model interpretability for healthcare applications.

Technical Skills

AI / LLM Engineering: LLMs (Claude, OpenAI, Gemini, Ollama), Prompt Engineering, RAG Pipelines, LangChain, AI Agents, Fine-tuning AI models
Computer Vision: YOLOv8, OpenCV, BLIP, CLIP, CNNs, Object Detection, Image Classification
ML / DL Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras, Pandas, NumPy
Backend & APIs: Python, FastAPI, REST APIs, MySQL, MongoDB, SQLite
Frontend: React, JavaScript, HTML5, CSS3, TailwindCSS, Vite
Tools & DevOps: Git, GitHub, Docker (basics), Postman, VS Code, Google Colab, MS Office

Education

Amity University Haryana <i>Major :B.Tech in CSE — Minor: Economics</i>	Gurugram, Haryana <i>2022 – 2026 — 8.12 CGPA</i>
Vaish Public School <i>Class XII, CBSE</i>	Rohtak, Haryana <i>2021 — 90.8%</i>
Vaish Public School <i>Class X, CBSE</i>	Rohtak, Haryana <i>2019 — 94.6%</i>

Achievements & Certifications

MLSA Beta: Microsoft Learn Student Ambassador (Beta) by Microsoft
President: MLSAxAUH Tech Society, Amity University Haryana
Hackathons: Top 10 at Hackverse, IILM University Haryana | Participated among 100 teams and got into finals at MasterX Hackathon — Participated in Triwizardathon and got into top 50 qualifying teams
Social Impact: Launched fundraising campaign for Pawzz animal welfare organization

Certifications

- ▷ SWAYAM – Python Programming
- ▷ NPTEL – Foundations of Virtual Reality & Augmented Reality
- ▷ McKinsey Forward Graduate Program